

Tandem Discharge

Turnkey Lovable Implementation Guide & Architecture Blueprint

Complete technical specifications, UI wireframes, TypeScript interfaces, mock NHS clinical cases, and modular copy-paste prompts to build a winning prototype in under 30 minutes.



Target Generator:	Lovable.dev / Claude Code / Vite+React	Design System:	Tailwind CSS + Tandem Teal Theme
Component Kit:	Radix UI / Shadcn UI + Lucide Icons	Build Duration:	30 - 45 min scaffold target
EHR Targets:	EMIS Web, SystmOne, Epic Hyperspace	Mock Datasets:	3 Pre-loaded NHS Inpatient/Urgent Care Cases
Status:	PRODUCTION READY PROMPTS	Offline Fallback:	100% resilient mock mode included

How to use this document with Lovable:

This functional specification is engineered for direct execution in **Lovable** (lovable.dev). It provides exact screen layouts, component trees, TypeScript data models, and a sequence of **5 copy-paste Master Prompts**. Feed the prompts sequentially to Lovable to generate the application rapidly, ensuring seamless UI polish, zero compile bugs, and instant live demo capabilities.

1. Application Architecture & UI Layout

The web application utilizes a split-screen **Dual-Pane Clinical Cockpit** layout. The left pane presents the raw, chronological patient record, while the right pane renders the AI-synthesized NHS Electronic Discharge Notification (eDN) across four specialized tabs.

Screen Region	Width / Grid	Components & Interactive Controls
Global Header	Full Width (h-16)	- Tandem Health Teal Logo + 'Discharge' badge. - Ward Selector: 'Ward 4B - Acute Medicine'. - Clinician Profile Pill: 'Dr. Alex Smith (FY1)'. Case Selector Dropdown: Toggle between Case 1 (Heart Failure), Case 2 (Urgent Care Asthma), Case 3 (Post-Op Cholecystectomy). 'Reset Demo' and 'New Synthesis' CTA buttons.
Left Pane: Inpatient Timeline Feed	38% Width (w-5/12)	Patient Demographic Banner: Name, NHS Number, Age, Bed #, Allergy status (red alert for Penicillin). Chronological Event Stack: - Day 0: Admission Clerking & Presenting Complaint. - Day 1: Consultant Ward Round & Bedside Echo. - Day 2: Blood Test Panel (U&Es; with eGFR drop, highlighted). - Day 3: Discharge Planning & Medication Adjustments. - Drug Chart: Inpatient Kardex snapshot. Paste Custom Notes: Accordion to paste raw text.
Right Pane: Synthesized eDN Cockpit	62% Width (w-7/12)	Top Command Bar: - Verification status badge ('NHS eDN Verified Draft'). - 'Show Source Citations' toggle. - 'Copy to EHR' modal trigger (EMIS / SystmOne / Epic). - 'Sign Off & Dispatch' button (triggers celebration). Four Interactive Tabs: 1. Medical eDN: Structured PRSB summary, SNOMED badges. 2. Med Rec Matrix: Color-coded 4-category diff table. 3. GP Action Plan: Prioritized checklist with timelines. 4. Patient Leaflet: Plain-English take-home guide.

1.2 Design System Tokens & Aesthetics

- **Primary Colors:** Teal 700 (#0F766E), Teal 600 (#0D9488), Teal 50 (#F0FDEA) matching Tandem Health's corporate identity.
- **Neutral Foundation:** Slate 900 (#0F172A) for headings, Slate 700 (#334155) for body, Slate 50 (#F8F8F8) for card backgrounds.
- **Clinical Category Accents:**
 - **Started:** Emerald (bg-emerald-50, text-emerald-700, border-emerald-300)
 - **Stopped:** Rose (bg-rose-50, text-rose-700, border-rose-300)
 - **Dose Changed:** Amber (bg-amber-50, text-amber-700, border-amber-300)
 - **Continued:** Blue (bg-blue-50, text-blue-700, border-blue-300)

2. TypeScript Core Data Models

Define these exact interfaces in `/src/types/clinical.ts` to ensure complete type safety:

```
export interface PatientDemographics {
  id: string;
  name: string;
  nhsNumber: string;
  dob: string;
  age: number;
  gender: string;
  bed: string;
  ward: string;
  admissionDate: string;
  dischargeDate: string;
  consultant: string;
  allergies: string[];
}

export interface TimelineEvent {
  id: string;
  dayNumber: number;
  date: string;
  time: string;
  category: 'admission' | 'ward_round' | 'labs' | 'microbiology' | 'radiology' | 'medication';
  author: string; // e.g. "Dr. A. Smith, FY1" or "Staff Nurse K. Miller"
  title: string;
  content: string;
  keyFindings?: string[];
  abnormalFlags?: string[];
}

export interface MedicationDiffItem {
  id: string;
  drugName: string;
  status: 'started' | 'stopped' | 'changed' | 'continued';
  priorDose?: string;
  dischargeDose: string;
  route: string;
  frequency: string;
  indication: string;
  clinicalRationale: string;
  plannedDuration?: string;
  gpInstructions?: string;
}

export interface GPActionItem {
  id: string;
  urgency: 'immediate_24h' | 'urgent_7d' | 'routine_4w' | 'safety_net';
  action: string;
  rationale: string;
  timeline: string;
  assignedRole: string; // e.g. "Practice Nurse", "GP", "Phlebotomy"
  completed: boolean;
}

export interface PatientLeafletData {
  reasonForAdmission: string;
  whatWeDid: string;
  currentCondition: string;
  dailyMedicineSchedule: {
    timeOfDay: 'Morning' | 'Lunch' | 'Evening' | 'Bedtime';
    medicines: string[];
    instructions: string;
  }[];
  redFlagSymptoms: string[];
  urgentContactInstructions: string;
}
```

3. Pre-loaded NHS Clinical Cases (Mock Data)

Embed these three realistic NHS clinical scenarios in /src/data/mockCases.ts. Case 2 is specifically tailored to appeal to Dr. Jonathan Tai's Urgent Care leadership.

Case Identifier	Clinical Specialty	Patient Narrative & Admin Friction	Key Synthesis & Med Rec Moments
Case 1: Mrs. Margaret Evans (82yo Female)	Acute Medicine / Frailty	Admitted with severe SOB, orthopnoea, and bilateral ankle oedema. Diagnosed with acute decompensated heart failure. Stayed 4 days on Ward 4B.	• Furosemide: Doubled to 80mg OD. • Ramipril: Withheld due to acute eGFR drop (62 -> 38). • GP Action: Check U&Es; in 7 days; restart ACEi when stable.
Case 2: Ms. Priya Sharma (28yo Female) <i>*Urgent Care Lead Special*</i>	Urgent Care / Ambulatory Acute	Severe acute asthma exacerbation triggered by viral URTI. Peak flow 45% predicted. Received 3x Salbutamol/Ipratropium nebs and IV Hydrocortisone in Urgent Care.	• Prednisolone: 40mg OD for 5 days started. • Inhaler Technique: Spacer provided. • GP Action: Mandatory asthma review within 48h; check peak flow diary.
Case 3: Mr. David Chen (45yo Male)	General Surgery / Post-Op	Elective laparoscopic cholecystectomy converted to mini-laparotomy. Developed post-op superficial surgical site infection with fever on Day 2.	• Co-Amoxiclav: 625mg TDS for 7 days. • Codeine: Ceased due to severe nausea/constipation; switched to Paracetamol. • GP Action: Practice nurse wound inspection at Day 7.

3.2 Realistic NHS Clinical Terminology Guardrails

- **Acronyms:** Use authentic NHS shorthand: SOB (Shortness of Breath), MFFD (Medically Fit for Discharge), U&Es; (Urea & Electrolytes), eGFR (estimated Glomerular Filtration Rate), TTO (To-Take-Out).
- **Safety Netting:** Enforce red flag triggers: chest pain, haemoptysis, acute breathlessness, signs of DVT/PE, sepsis warning signs (NEWS2 score triggers).

4. Lovable Master Prompts — Phase 1 & 2

Copy and paste Prompt 1 into Lovable to scaffold the foundation. Once complete, submit Prompt 2.

Prompt 1: Project Shell, Theme & Layout Scaffold

Build a modern clinical healthcare web app called 'Tandem Discharge'.
Theme: Clean, sleek medical design inspired by Tandem Health. Primary color: Teal (#0F766E, #0D9488), background: Slate-50 (#F8FAFC), card borders: Slate-200, dark slate text.
Header: Left side has a brand icon with text 'Tandem Discharge' and a subtle badge 'NHS eDN Synthesizer'. Center has a Ward Selector pill ('Ward 4B - Acute Medicine') and Clinician profile ('Dr. Alex Smith, FY1'). Right side has a Case Selector Dropdown ('Case 1: Heart Failure (82yo F)', 'Case 2: Urgent Care Asthma (28yo F)', 'Case 3: Post-Op Cholecystectomy (45yo M)') and a 'Reset Demo' button.
Main layout: Responsive 2-column split screen (Left: 38% width 'Inpatient Timeline Feed', Right: 62% width 'Synthesized eDN Cockpit'). Use Lucide icons throughout.

Prompt 2: Inpatient Timeline Feed & Demographics Component

In the Left Pane ('Inpatient Timeline Feed'), build a rich chronological record viewer:

1. Patient Demographic Banner at top: Name, NHS Number (xxx xxx xxxx), Age, Bed, Admission Date, Consultant, and an Allergy badge (Show 'NKDA' or red warning 'ALLERGY: Penicillin').
2. Chronological Vertical Timeline: Render event cards along a vertical teal line for Day 0 (Admission Clerking), Day 1 (Consultant Ward Round), Day 2 (Blood Tests / Labs), Day 3 (Discharge Decision), and Inpatient Drug Chart. Each card has a date badge, author role, title, and bulleted findings.
3. Highlight abnormal lab values with an amber pill (e.g. 'eGFR: 38 mL/min [Abnormal Drop]', 'CRP: 84 mg/L').
4. Include an 'Add Ward Note' accordion at the bottom allowing custom text pasting.

5. Lovable Master Prompts — Phase 3 & 4

Execute Prompt 3 to build the tabbed eDN workspace, followed by Prompt 4 for the Med Rec & GP Checklist.

Prompt 3: Synthesized eDN Cockpit & 4 Interactive Tabs

In the Right Pane, build the 'Synthesized eDN Cockpit':

1. Top Command Bar: Status badge ('Verified NHS eDN Draft'), 'Source Tracing' toggle switch, 'Copy to EHR' button with clipboard icon, and 'Sign Off & Dispatch' primary teal button.
 2. Four primary Tabs using Radix UI:
 - Tab 1: 'Medical eDN (PRSB)' - Formatted according to NHS PRSB standards: Primary Diagnosis with SNOMED-CT code badge, Secondary Diagnoses, Clinical Narrative / Hospital Course, Procedures, Complications, and Discharge Destination.
 - Tab 2: 'Medication Reconciliation' - A visual diff table comparing pre-admission vs discharge meds.
 - Tab 3: 'GP Action Plan' - Categorized checklist of GP follow-up tasks.
 - Tab 4: 'Patient Take-Home Leaflet' - Plain-English summary for the patient.
- Make all fields editable so the clinician has full human-in-the-loop control.

Prompt 4: Medication Reconciliation Diff Matrix & GP Action Checklist

Implement Tab 2 (Medication Reconciliation) and Tab 3 (GP Action Plan):

For Tab 2: Build a table with columns [Medication Name, Status, Prior Dose, Discharge Dose, Route/Freq, Indication, Clinical Rationale for Change]. Render status as badges:

- Green: STARTED (e.g. Furosemide 40mg IV)
- Red: STOPPED (e.g. Ramipril 5mg OD - Rationale: 'Withheld due to acute eGFR decline from 62 to 38')
- Amber: DOSE CHANGED (e.g. Furosemide increased to 80mg PO OD)
- Blue: CONTINUED (e.g. Atorvastatin 20mg ON)

For Tab 3: Build an interactive checklist grouped by urgency: 'Immediate (24-48h)', 'Urgent (7 Days)', and 'Routine (2-4 Weeks)'. Each item has a checkbox, bold action title, clinical rationale, and assignee pill (e.g. 'Phlebotomy / Practice Nurse: Repeat U&Es; in 7 days to reassess renal function prior to restarting ACEi').

6. Lovable Prompt 5 & Saturday Demo Checklist

Execute Prompt 5 for final visual polish, citation highlights, and export modals:

Prompt 5: Patient Leaflet, Source Citations, EHR Modal & Confetti Polish

Add the finishing wow-factor features:

1. Tab 4 ('Patient Leaflet'): Friendly card layout. 'Why you were in hospital' in simple language (Reading age 11). A 4-column visual table for 'Your Daily Medicines' (Morning, Noon, Evening, Bedtime). A prominent alert card: 'When to get urgent medical advice' with 111 vs 999 rules.
2. Source Tracing: When the 'Source Tracing' toggle is on, clicking or hovering over any synthesized section highlights the corresponding event card in the Left Timeline with a pulse animation and teal border.
3. EHR Export Modal: Clicking 'Copy to EHR' opens a modal with tabs for 'EMIS Web', 'SystemOne', and 'Epic'. Each tab shows formatted text ready for one-click copy, with a toast notification 'Copied to clipboard formatted for EMIS!'.
4. Sign Off & Dispatch: Clicking this button fires canvas-confetti and shows a success modal: 'Discharge Summary Dispatched to Millwood Medical Practice & NHS Spine'.

6.2 Saturday Hackathon Demo Checklist (15:55 Live Pitch)

- **Display Pre-Flight:** Connect laptop to presentation display; set browser zoom to 90% for optimal dual-pane visibility.
- **Case 1 Warm-up:** Pre-load Mrs. Margaret Evans (Heart Failure). Point out the chaotic 4-day ward notes on the left.
- **The Synthesis Reveal:** Demonstrate the 4-color Med Rec diff table. Explain why Ramipril was stopped (eGFR drop).
- **The Urgent Care Appeal:** Briefly switch to Case 2 (Ms. Priya Sharma, Asthma) to directly appeal to Dr. Jonathan Tai.
- **The Grand Finale:** Toggle to the Patient Leaflet, click 'Copy to EHR (EMIS)', and hit 'Sign Off & Dispatch' to trigger the confetti celebration.